

APPLICATIONS OF DIGITAL TWINS IN PREDICTIVE MAINTENANCE OF MECHANICAL SYSTEMS

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ABSTRACT

Digital twins are revolutionizing predictive maintenance by providing real-time virtual replicas of mechanical systems. These models enable continuous monitoring, early fault detection, and predictive analysis, allowing maintenance teams to foresee and prevent equipment failures. By simulating different maintenance scenarios and forecasting component lifespan, digital twins optimize maintenance planning and support remote diagnostics. When integrated with IoT and machine learning, this technology enhances reliability, reduces downtime, and lowers maintenance costs. This paper examines the key applications of digital twins in predictive maintenance, focusing on their role in improving efficiency and extending the lifespan of mechanical systems.

Keywords: digital twins, predictive maintenance, mechanical systems, real-time monitoring, fault detection, PdMDT

1. DEFINITION OF DIGITAL TWIN

The increasing complexity of mechanical systems in industries such as manufacturing, aerospace, and automotive has amplified the need for more efficient and reliable maintenance strategies. Traditional maintenance approaches, often relying on reactive or scheduled interventions, are becoming insufficient in dealing with the growing demand for uptime, performance, and cost optimization. Predictive maintenance offers a promising solution by forecasting potential failures and enabling timely interventions before critical issues arise.

One of the key enablers of predictive maintenance is the concept of the Digital Twin (DT). By integrating digital twins with advanced analytics, artificial intelligence (AI), and machine learning (ML) techniques, it becomes possible to predict failures, optimize maintenance schedules, and improve overall system performance [1].

Digital twins, considered a key enabling technology, serve as virtual models of physical objects, processes, and real-time data throughout the entire product life cycle. The idea of digital twins was first introduced by Michael Grieves in 2002 during a discussion on Product Lifecycle Management (PLM). In 2012, Grieves and John Vickers further developed the concept, defining digital twins as digital replicas of physical assets that maintain an automatic and continuous connection between the physical and digital realms [2].

The core functionality of a digital twin is based on three primary components, as shown in Figure 1. These components include: the physical entity, the digital or virtual counterpart, and the connection between the physical and digital entities [2-5].

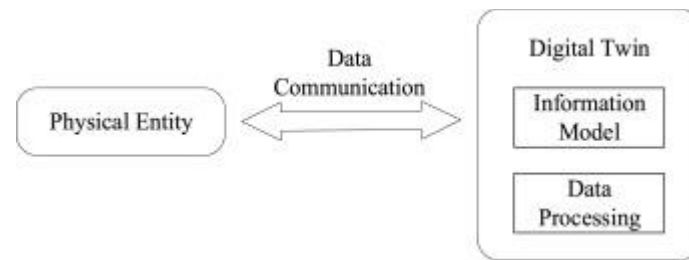


Figure 1. General reference model of digital twin [2]

1.1. Historical development of digital twins

The concept of digital twins has its origins in the early 2000s. It has evolved significantly over the years, driven by advancements in digital technologies, computing power, and data analytics. As previously mentioned, the term "digital twin" was first introduced by Michael Grieves in 2002. Grieves proposed the idea of a digital twin as a virtual model of a product or system, designed to improve product design and manufacturing. His early concept aimed at creating a digital counterpart of a physical asset that would follow it through its entire lifecycle, from design to decommissioning [6]. In these early days, digital twins were primarily used to enhance product design through simulation and virtual prototyping.

The development of the Internet of Things (IoT) and advancements in data analytics during the 2010s accelerated the adoption of digital twins across multiple industries. The ability to connect physical systems to the internet allowed for more extensive data collection, making it feasible to create highly accurate and dynamic digital twins. This period saw digital twins being applied not only in product design and manufacturing but also in operations, predictive maintenance, and asset management across industries like aerospace, automotive, and energy sectors [3].

In recent years, the applications of digital twins have expanded into numerous sectors, including healthcare, construction, urban planning, and smart cities. For instance, in healthcare, digital twins are being used to model patient-specific treatments by creating virtual models of organs or even entire physiological systems. In the energy sector, digital twins are utilized to monitor and optimize the performance of power grids and renewable energy systems [7]. Moreover, the incorporation of artificial intelligence (AI) and machine learning (ML) into digital twin frameworks has enabled predictive analytics, allowing organizations to forecast failures or inefficiencies before they occur and make proactive adjustments to improve performance and reduce costs [5].

2. DIGITAL TWINS AND PREDICTIVE MAINTENANCE

Although digital twins have significantly advanced in recent years, their application in design was not fully explored until the study by [8]. This research introduced a catalogue of digital twin design patterns. The authors proposed specific patterns for different stages of the product development process and clarified what does not qualify as a digital twin, such as a digital model, digital generator, or digital shadow. According to their view, a true digital twin consists of a causally connected and synchronized digital and physical object.

2.1. Maintenance

Maintenance is an essential part of ensuring the reliability, efficiency, and safety of mechanical systems. Various maintenance strategies are employed depending on the complexity of the

systems, operational requirements, and associated costs. The main five types of maintenance include the following: reactive maintenance, preventive maintenance, condition-based maintenance, predictive maintenance and prescriptive maintenance [1,9].

Predictive maintenance focuses on proactively reducing costs and improving machine uptime by predicting when a component or system might fail. It estimates when a part will no longer function effectively, typically measured through indicators like remaining useful life (RUL), health metrics, or equipment efficiency. Additionally, this approach promotes sustainability by fully utilizing assets without causing wear on other components or wasting resources [9].

2.2. Characteristics of predictive maintenance based on digital twin (PdMDT)

PdMDT has several defining characteristics that distinguish it from traditional maintenance practices, combining the predictive analytics of condition-based maintenance with the comprehensive virtual modelling of a system's digital twin.

3.2.1. Real-time perception

One of the key characteristics of PdMDT is the continuous monitoring of physical assets through real-time data collection. This data is collected via embedded sensors and IoT (Internet of Things) devices that track various operational parameters such as temperature, vibration, pressure, and load.

The real-time data is fed directly into the digital twin model, which provides a dynamic, virtual representation of the physical system. This integration enables the digital twin to reflect the current state of the equipment accurately, allowing it to predict failures based on actual operating conditions, rather than fixed schedules or average historical data [10].

3.2.2. Predictive analytics and machine learning

A defining feature of PdMDT is its ability to predict future failures and maintenance needs using predictive analytics. By analysing data trends, machine learning algorithms can forecast when a component is likely to fail and calculate the remaining useful life (RUL) of the system's parts.

The digital twin leverages historical data, real-time inputs, and operational simulations to identify patterns and anomalies that could signal an impending failure. This proactive approach allows maintenance teams to address issues before they cause downtime or damage [11].

Furthermore, machine learning models continuously learn and improve over time, refining their ability to predict failures with greater accuracy as they analyse more data.

3.2.3. Virtual simulation and optimisation

Digital twins in predictive maintenance allow for virtual simulation of mechanical systems under various operational conditions. This means that maintenance teams can simulate different scenarios, such as increased load or fluctuating temperatures, to understand how these conditions affect system performance and longevity.

Through “what-if” simulations, different maintenance strategies can be tested without risking the actual equipment. Engineers can evaluate potential outcomes and optimize their approach to avoid costly repairs and minimize downtime [12].

Beside these simulations, the digital twin can also iterate maintenance schedules, suggesting the most effective intervention times to ensure the machinery operates at peak performance while avoiding unnecessary maintenance [13].

3.2.4. Two-way communication between physical and digital systems

PdMDT creates a closed-loop system where real-time data flows continuously between the physical asset and its digital twin. This two-way communication enables the digital twin to constantly update and adjust based on the actual conditions of the mechanical system. For example, changes in the physical asset, such as wear or stress, are mirrored in the digital twin, while the insights generated by the twin (e.g., predicted failure times) are relayed back to the maintenance team. This synchronization ensures that the virtual model always represents the current state of the physical system [14].

2.3. Difference between PdMDT and traditional maintenance

Predictive maintenance with digital twins (PdMDT) represents a significant advancement in maintenance strategies compared to traditional maintenance approaches. The use of digital twins provides an enhanced layer of functionality, allowing more precise and effective maintenance scheduling and optimization. The main differences between PdMDT and traditional maintenance are summarized in Table 2.

Table 2. Summary of differences between PdMDT and traditional maintenance

<i>Aspect</i>	<i>Traditional maintenance (reactive/preventive)</i>	<i>Predictive maintenance with digital twins (PdMDT)</i>
<i>Approach</i>	<i>Reactive: after failure Preventive: time-based</i>	<i>Real-time, condition-based maintenance</i>
<i>Data utilization</i>	<i>Minimal or based on historical data [2]</i>	<i>Extensive real-time data with IoT and sensors [2]</i>
<i>Predictive capability</i>	<i>None or limited</i>	<i>Advanced predictions using machine learning and simulations</i>
<i>Downtime</i>	<i>High in reactive; moderate in preventive</i>	<i>Minimal downtime due to optimised scheduling [12]</i>
<i>Cost efficiency</i>	<i>Higher long-term costs due to breakdowns or over-maintenance</i>	<i>More cost-efficient, reduces unplanned maintenance [4]</i>
<i>Technological investment</i>	<i>Low, simple implementation</i>	<i>High, requires digital twins, sensors, data analytics</i>

3. APPLICATIONS OF PdMDT

Predictive maintenance based on digital twin (PdMDT) has revolutionized the management of mechanical systems, offering predictive insights into the health of machinery and enabling real-time, data-driven maintenance decisions. By integrating IoT (Internet of Things), sensors, and simulation models, PdMDT provides a robust framework for monitoring the operational status of mechanical systems. This integration helps reduce unplanned downtime, improve performance, and extend the life of equipment. PdMDT finds applications in various industries that rely heavily on mechanical systems, such as manufacturing, energy, aerospace, and automotive sectors.

In manufacturing, PdMDT plays a critical role in monitoring production machinery, such as CNC (computer numerical control) machines, hydraulic presses, and robotic systems. These machines are often subjected to constant use, and failures can lead to significant production losses. By using PdMDT, manufacturers can continuously assess the wear and tear of components such as bearings, motors, and gears, predicting when they will fail based on real-time operational data.

In the automotive industry, PdMDT is applied in the maintenance of vehicle fleets and manufacturing equipment. Automotive companies use digital twins to monitor the health of robotic arms in assembly lines, as well as to track the performance of electric vehicles (EVs) in real time. For instance, digital twins are used to monitor the battery life and charging cycles

of EVs, allowing manufacturers and fleet operators to predict when a battery might fail or require maintenance.

PdMDT is also applied in the maintenance of railway systems, where it helps monitor the condition of trains and rail infrastructure. Railway systems are complex, with multiple components such as locomotives, tracks, and signalling systems requiring regular maintenance to ensure safe and efficient operation.

Many papers and articles on PdMDT applications in mechanical systems are found and some of them are described in the following sections.

3.1. Prediction of the RUL of an industrial robot

The paper [15] introduces a methodology for calculating the remaining useful life (RUL) of industrial machinery by utilizing the digital twin concept. The focus is on predictive maintenance within the context of manufacturing, where the goal is to avoid unexpected machine failures and optimize maintenance schedules.

The case study described in the paper involves a six-axis industrial robot, used for welding tasks in the assembly of a thermosyphonic system. The robot's gearboxes were chosen as the focus for predictive maintenance due to their critical role in the robot's functionality. The digital twin of the robot, Figure 2, was created using physics-based simulations and real-time data gathered from the robot's controllers and external sensors. The data was used to tune the digital model continuously, ensuring it accurately reflected the condition of the real robot.

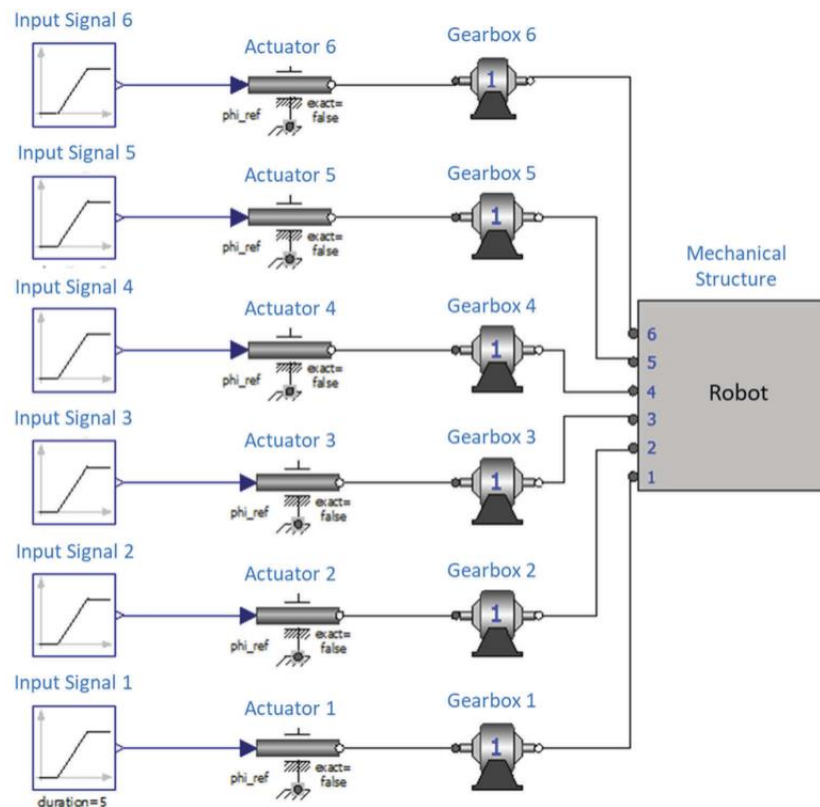


Figure 2. Robot digital model [15]

The prediction of the RUL for the industrial robot is primarily based on a combination of real-time data and physics-based simulation models. The digital twin of the robot continuously receives data from the machine's controller and sensors, including torque and position signals from the gearboxes. These signals are compared to the outputs from the simulation model, which predicts the expected torque values.

The RUL estimation for each gearbox was performed once per day during the case study, allowing maintenance schedules to be adjusted in real-time based on the current state of the robot. This approach enables the proactive scheduling of maintenance activities, avoiding unexpected breakdowns and extending the lifespan of the robot's components.

3.2. Design of a predictive maintenance algorithm for a triplex pump

An article [16] explains how a predictive maintenance algorithm, Figure 3, for a triplex pump was developed using MATLAB®, Simulink, and Simscape™. A digital twin of the pump was created in Simscape™, fine-tuned with real-world data, and used in conjunction with machine learning to build the predictive maintenance system. The algorithm relies solely on the pump's outlet pressure to detect which parts, or combinations of parts, are likely to fail, providing an efficient way to forecast maintenance needs.

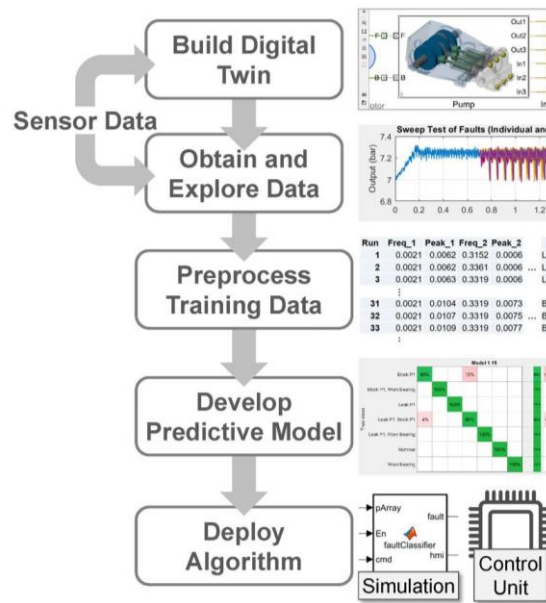


Figure 3. Predictive maintenance algorithm for a triplex pump [16]

By creating a physics-based simulation of the pump and combining it with real-time sensor data like pressure and vibration, the system can monitor the pump's health and simulate various fault conditions. Machine learning algorithms are applied to analyze this data, predict potential failures in components such as plungers and valves, and schedule maintenance proactively. This approach helps minimize downtime and optimize repair schedules.

The digital twin thus provides a proactive maintenance system, allowing for faults to be detected early and interventions to be scheduled based on data-driven insights.

3.3. Digital twin of an automotive brake pad

The development of a digital twin model used to predict the wear of automotive brake pads is presented in [17]. This digital twin utilizes real-time brake pressure data collected through the ThingWorx Internet of Things (IoT) platform and is simulated in CREO Simulate for predictive maintenance purposes. The authors focus on simulating the brake pad and rotor assembly under varying vehicle speeds and brake pressures to determine wear rates. By comparing calculated and simulated data, the results show good correlation with an average error of 11%.

The methodology involves capturing brake pressure data at different vehicle speeds using sensors and feeding this data into the CREO model for simulation, Figure 4. Brake pad wear is calculated using analytical equations, showing that wear increases with both speed and

pressure. This approach demonstrates how real-time data combined with simulation tools like CREO can help predict brake pad wear and optimize maintenance schedules.

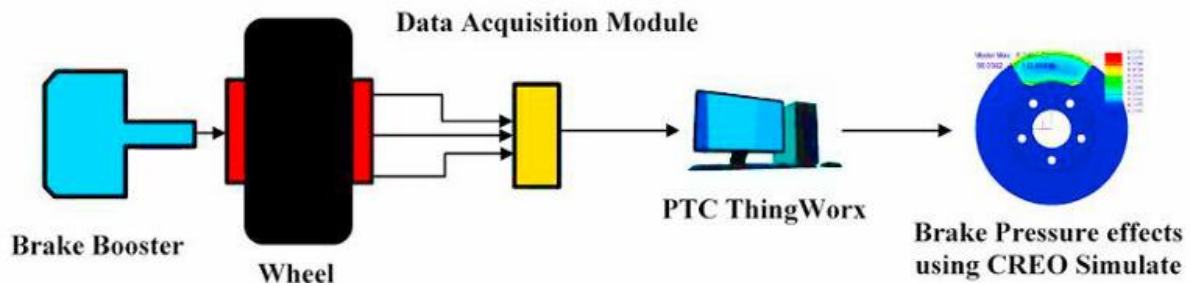


Figure 4. The process flow of the proposed predictive maintenance system [17]

The authors highlight that automotive manufacturers have yet to widely adopt advanced predictive maintenance solutions for braking systems. However, they foresee the potential of combining machine learning and digital twin technologies to create more accurate predictive models for brake systems. Future work includes expanding this proof of concept to a full vehicle digital twin, which could revolutionize maintenance practices in the automotive industry.

3.4. Digital twin for machining tool condition prediction

The paper [18] focuses on utilizing digital twins for predicting the condition of machining tools and performing predictive maintenance. Digital twin technology integrates real-time data, simulations, and advanced algorithms to monitor tool wear and performance degradation in machining systems. By combining physical models, virtual sensors, and machine learning techniques like deep learning, this approach enhances the accuracy of fault prediction in tools, leading to more efficient maintenance and operation.

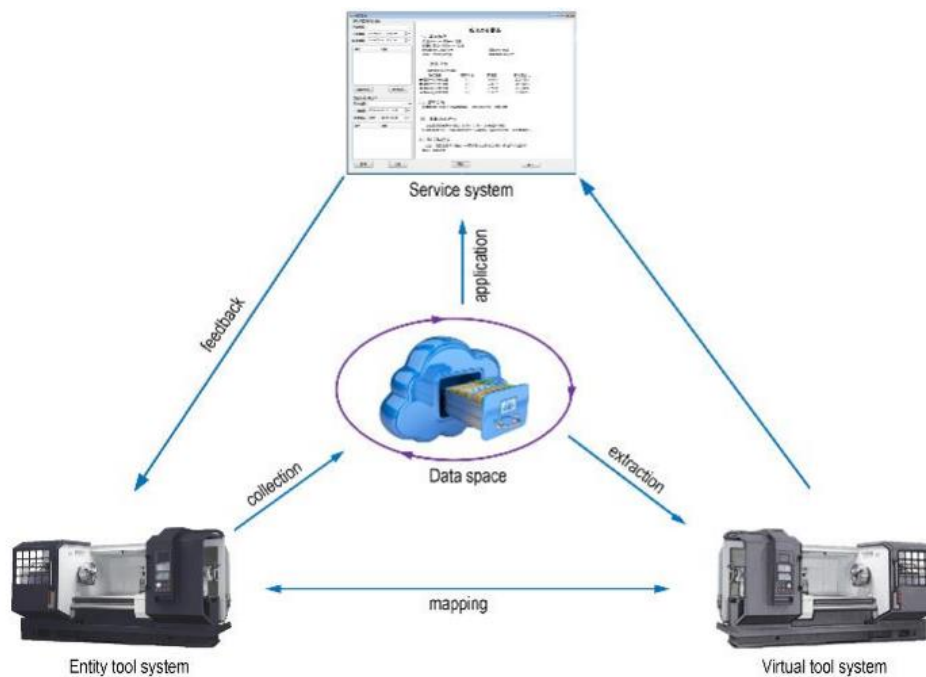


Figure 5. Digital Twin model of tool system [18]

The methodology includes creating a five-dimensional digital twin model that incorporates both physical and virtual representations of the tool system, Figure 5. Deep learning techniques,

particularly the Deep Stacked Gated Recurrent Unit (DSGRU), are applied to the digital twin model for tool wear prediction. This model improves tool condition monitoring by analysing real-time sensor data, such as vibrations and cutting forces, from machining processes.

In the experimental setup, a CNC machine tool was monitored using sensors to collect data on vibrations and cutting forces in three axes (X, Y, Z). This data was then processed using deep learning models to predict tool wear and detect potential faults. The results showed that the digital twin model, combined with the deep learning approach, significantly improved the accuracy of tool wear predictions, enabling better maintenance planning and reduced downtime.

The authors conclude that digital twin technology, supported by advanced data analytics and machine learning, offers a promising solution for predictive maintenance in machining systems. It allows for real-time monitoring and proactive fault detection, which can extend the life of tools and improve overall operational efficiency in manufacturing environments.

3.5. Digital twin for predictive maintenance of gearboxes in wind turbine drivetrains

The development of a digital twin model for predicting the remaining useful life (RUL) of drivetrain components in floating offshore wind turbines (FWTs) is explored in [19]. These systems are particularly prone to failures due to the dynamic environment they operate in, experiencing a combination of wind, wake, and wave forces that make maintenance and failure prevention critical. The drivetrain system, which includes the rotor, bearings, gearbox, and generator, contributes to a significant portion of turbine failures, and predictive maintenance has the potential to reduce unexpected shutdowns and optimize maintenance schedules.

The digital twin concept introduced in this paper is based on a multi-degree-of-freedom torsional model that mimics the real-time behaviour of the drivetrain system. By analysing torsional vibration measurements, this digital twin model can predict the degradation of gearbox components, such as gears, which are vulnerable to contact fatigue damage.

To demonstrate the application of this approach, the authors conducted simulations using a 5MW reference floating turbine model. The digital twin model estimates dynamic properties like natural frequencies and damping, which are key indicators of drivetrain health. As gears degrade due to fatigue, these dynamic properties shift, and the model can track these changes in near real-time to assess the remaining life of the drivetrain components. The algorithm proposed for parameter estimation is designed to be computationally fast, making it suitable for real-time applications, with minimal processing delays.

A key aspect of the model is its ability to estimate contact stresses between gear pairs and to calculate the accumulation of fatigue damage using a physics-based degradation model. The stress data is analysed using the rainflow-counting method to estimate the cumulative damage to gear teeth over time. Simulation results showed a strong correlation between the digital twin model's predictions and high-fidelity simulation models, confirming its accuracy in estimating gear degradation.

However, the practical implementation of such models comes with challenges. Uncertainties in real-time measurements and model estimation can affect performance, though the authors propose methods to mitigate these issues. The computational simplicity of the model ensures that it can be integrated into existing wind turbine monitoring systems with little additional cost.

In conclusion, the paper presents a promising approach to extending the operational life of wind turbines through predictive maintenance. By utilizing a digital twin model, wind farm operators can monitor the drivetrain's health in real-time, allowing for timely maintenance interventions that prevent costly turbine failures. The paper suggests future work in hardware-in-the-loop simulations to further validate the approach for real-time operational use.

4. CHALLENGES AND FUTURE PERSPECTIVES OF PdMDT

As described in previous sections, predictive maintenance with digital twins offers numerous advantages, including improved system reliability, cost efficiency, and enhanced decision-making. However, despite these advantages, PdMDT faces several challenges that hinder its widespread adoption. One of the primary challenges is the high initial investment required to implement the technology. Organizations must invest in infrastructure, software, and expertise to effectively deploy PdMDT. The cost of sensors, IoT devices, and data infrastructure needed to collect and process real-time data can be prohibitive, especially for smaller companies with limited budgets [11]. Additionally, developing accurate and functional digital twins is a complex process that requires specialized expertise in modelling, data science, and system integration, further contributing to the high upfront costs [12].

Another significant challenge associated with PdMDT is the management and integration of large amounts of data. The real-time data generated by sensors and IoT devices can be overwhelming, and organizations must have robust systems in place to collect, store, and analyse this data effectively.

Cybersecurity is also a concern for organizations adopting PdMDT. The reliance on IoT devices and cloud computing introduces vulnerabilities, as these systems can be potential targets for cyberattacks. IoT devices, in particular, are often less secure than traditional IT systems, making them susceptible to breaches that could compromise the integrity of maintenance data. Ensuring the security of data transmission between physical assets and their digital twins is critical to preventing data breaches and maintaining trust in the system [12].

Looking ahead, the future of PdMDT is promising, with numerous opportunities for further development. One of the most significant future trends is the integration of artificial intelligence and machine learning into PdMDT systems. These technologies will enable more advanced predictive capabilities and smarter decision-making processes.

In addition to AI integration, PdMDT is expected to expand into a wider range of industries and sectors. Currently, PdMDT is most commonly used in industries such as manufacturing and energy, where the benefits of predictive maintenance are well recognized [11]. However, as the cost of IoT devices and digital twin technology decreases, other sectors such as healthcare, transportation, and infrastructure management are likely to adopt PdMDT to improve the maintenance of their critical assets [12].

5. CONCLUSION

The integration of digital twin technology into predictive maintenance of mechanical systems represents a transformative advancement in industrial operations. By creating a virtual replica of physical assets, digital twins enable real-time monitoring, simulation, and analysis of system performance, leading to enhanced decision-making processes. The applications of digital twins extend across various sectors, including manufacturing, aerospace, automotive, and energy, demonstrating their versatility and effectiveness.

One of the primary advantages of employing digital twins in predictive maintenance is their ability to provide insights into equipment health and performance metrics. By leveraging data analytics and machine learning algorithms, organizations can identify potential failures before they occur, reducing unplanned downtime and minimizing maintenance costs. This proactive approach not only improves asset reliability but also extends the lifespan of mechanical systems, allowing for optimized resource allocation.

In conclusion, the application of digital twins in predictive maintenance is not merely a technological enhancement; it signifies a paradigm shift in how organizations approach asset management. By embracing this innovative approach, industries can achieve significant operational efficiencies, cost savings, and improved safety outcomes. As digital twin

technology continues to evolve and become more accessible, its role in predictive maintenance will undoubtedly expand, paving the way for smarter, more resilient mechanical systems in the future.

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